

Finite element method in a transient heat transfer and stress analysis

Mukundan Subbaiya

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1. Introduction

An important aspect in gas turbine machinery is the structural integrity of high-pressure turbine discs, and this is especially significant during transient start-up conditions. The present study aims to investigate the high-pressure turbine disc of a SIEMENS GT35C turbine, which experiences high centrifugal stresses due to its operating speed of 7000 RPM and significant thermal gradients due to hot gas flow effects.

The objective of the study is to monitor the evolution of the stress state from the cold-start condition to a constant operating temperature in 3000 seconds. To achieve this, we use a coupled thermo-mechanical analysis, where the temperature field is initially calculated by solving the transient heat transfer equation, and then used to calculate the quasi-static structural response. Of particular interest is the tangential (hoop) stress at the inner bore ($r = 0.075$ m).

2. Analytical Verification: Simplified Disc Model

Before attempting complex geometries, it is critical to validate the FE formulation against a reliable analytical solution. This process ensures that the software has correctly applied boundary conditions, material properties, and centrifugal loading.

2.1 Analytical Formulation and Benchmark Calculation

For a disc of constant thickness, the tangential stress σ_θ is derived from the radial equilibrium equation. The solution must account for two distinct mechanical loads: the centrifugal body force of the disc itself and the external radial pressure exerted by the turbine blades (blade pull). The combined stress is given by:

$$\sigma_\theta(r) = \underbrace{\frac{3+\nu}{8}\rho\omega^2\left(a^2+b^2+\frac{a^2b^2}{r^2}-\frac{1+3\nu}{3+\nu}r^2\right)}_{\text{Centrifugal Component}} + \underbrace{\frac{Pb^2}{b^2-a^2}\left(1+\frac{a^2}{r^2}\right)}_{\text{Blade Pull Component}} \quad (1)$$

Using the parameters $\rho = 7940$ kg/m³, $\nu = 0.3$, $a = 0.075$ m, $b = 0.4002$ m, and $\omega \approx 733.04$ rad/s, and a blade load $P = 140$ MPa:

- **Centrifugal Stress:** The rotation induces a peak hoop stress at the inner bore ($r = a$) due to the summation of radial mass acceleration. At $r = 0.075$, $\sigma_{\theta, \text{centrifugal}} \approx 568.2$ MPa.
- **Blade Pull Stress:** The blades act as a distributed radial load at the rim. At $r = 0.075$, $\sigma_{\theta, \text{blade}} = 140 \cdot \frac{0.4002^2}{0.4002^2 - 0.075^2} \cdot (1 + 1) \approx 290.1$ MPa.
- **Total Benchmark:** The total analytical stress is **858.3** MPa.

The numerical model for the constant thickness disc yielded a value of **848.22** MPa, representing a relative error of **1.17%**, confirming the accuracy of the FE setup.

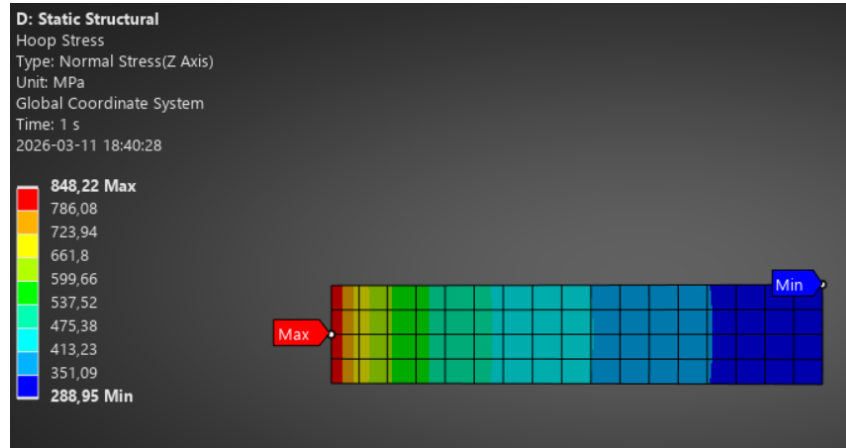


Figure 1: Tangential stress distribution in the constant-thickness verification model.

3. Methodology and Approximation Levels

Engineering simulations require a balance between computational efficiency and physical accuracy, a hierarchical modeling approach was adopted.

3.1 2D Axisymmetric Modeling

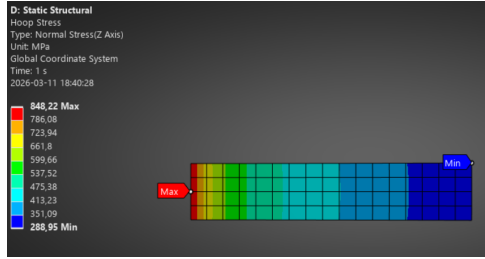
In view of this rotational symmetry, a two-dimensional axisymmetric analysis was used. This effectively reduces a three-dimensional analysis to a two-dimensional domain, significantly cutting down computation time. Axisymmetric analysis is quite effective in capturing radial and tangential stress conditions.

3.2 Geometric Refinement

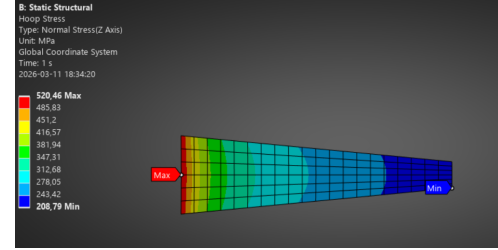
The analysis progressed through three stages:

1. **Constant Thickness:** Validated the mathematical foundations.
2. **Tapered Geometry:** To show that reducing the thickness results in modifying the stress flow.

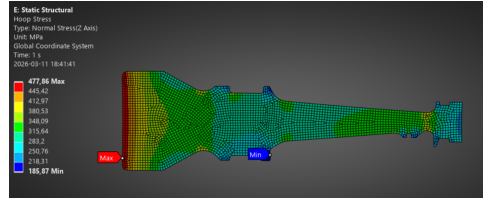
3. Detailed Geometry: The final model included fillets. These features are critical as they mitigate stress concentrations that are otherwise ignored in simplified models.



(a) Constant Thickness



(b) Tapered Geometry



(c) Detailed Geometry

Figure 2: Comparison of tangential (hoop) stress distributions across the three levels of geometric approximation at steady-state.

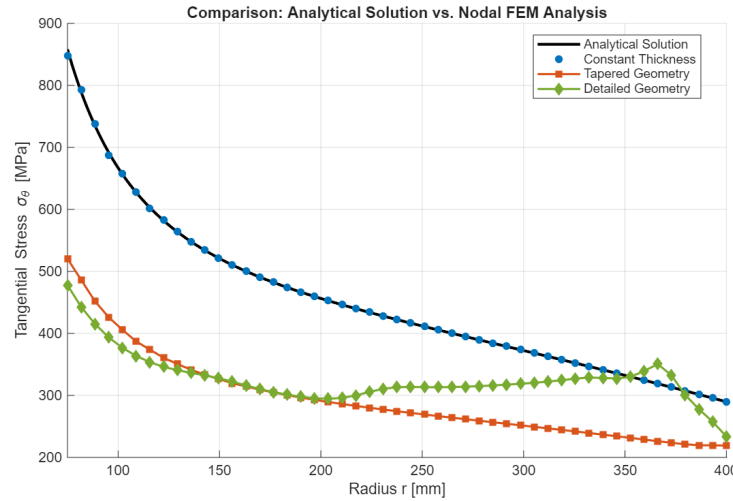


Figure 3: Comparison plot of Analytical Solution v/s FEM analysis with different models.

4. Steady-State Thermo-Mechanical Analysis

Following the model verification on mechanical load, the detailed geometry was subjected to steady-state thermal loads to evaluate the interaction between temperature gradients and centrifugal forces.

4.1 Steady-State Temperature Field

Thermal boundary conditions, defined by the convection coefficients (α) and ambient temperatures (T_∞) at full load ($t > 250$ s), were applied. The resulting temperature field shows a significant gradient between the inner and the outer radius. The outer radius reaches approximately of maximum temperature of **492.37 °C**, while the inner radius settles at **291.81 °C**.

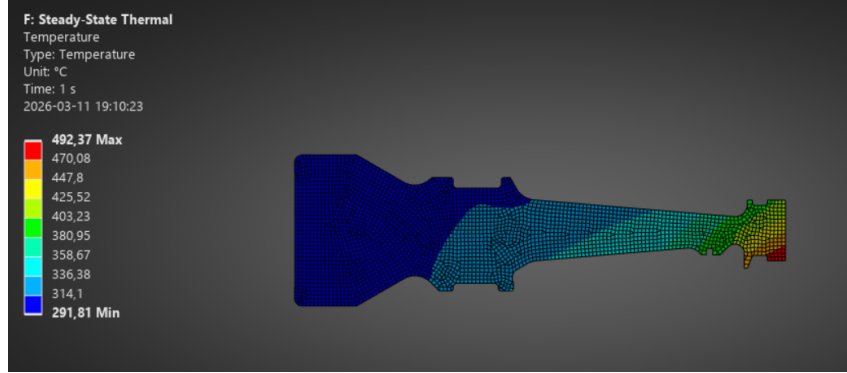


Figure 4: Steady-state temperature distribution (T) across the detailed disc geometry.

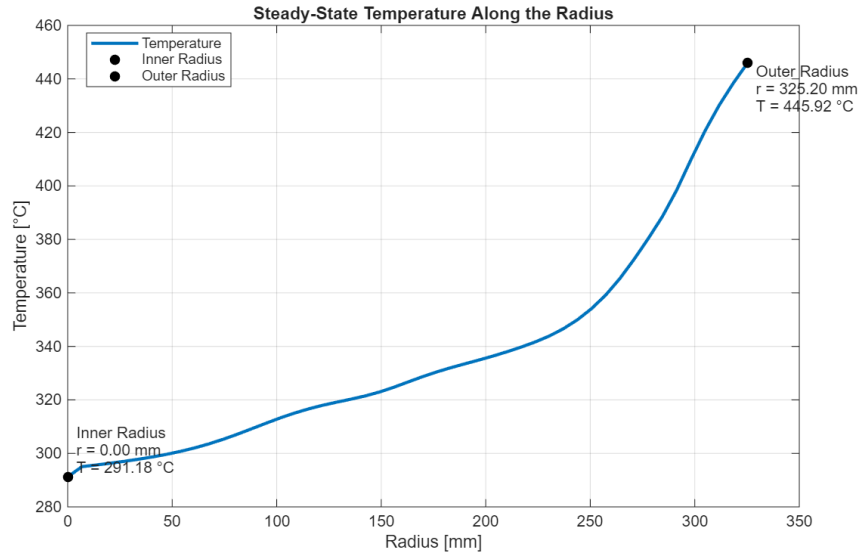


Figure 5: Temperature along the radial coordinate of the Disc.

4.2 Tangential Stresses: Thermal Loading Only

To isolate the effect of the temperature field, a structural analysis was performed with $\omega = 0$ (no rotation) and $P = 0$ (no blade pull). The results indicate that the temperature variation alone creates significant tangential tension at the inner radius and compression at the outer radius.

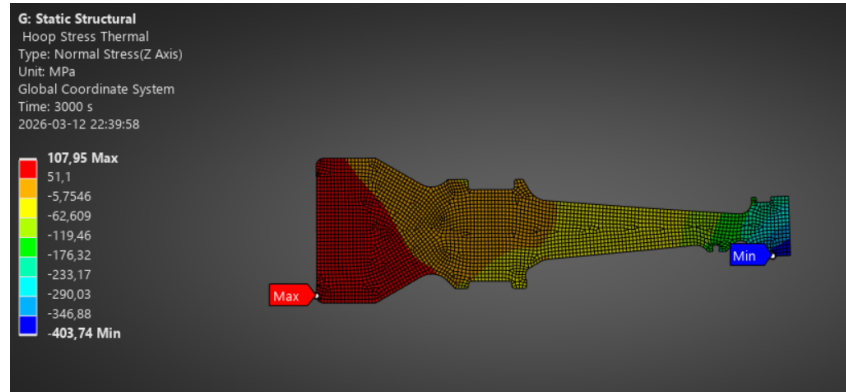


Figure 6: Tangential stresses due only to the temperature variations at steady state.

Comparison: Unlike the purely mechanical stress, which is driven by centrifugal mass, this thermal stress is driven by the "clamping" effect: the hotter outer material expands but is restricted by the cooler inner material. At $r = 0.075\text{m}$, the thermal-only hoop stress is **107.95 MPa**.

4.3 Combined Thermo-Mechanical Steady-State Stress

Finally, the full operational load was applied, combining the 7000 RPM centrifugal force, the blade pull, and the steady-state temperature field.

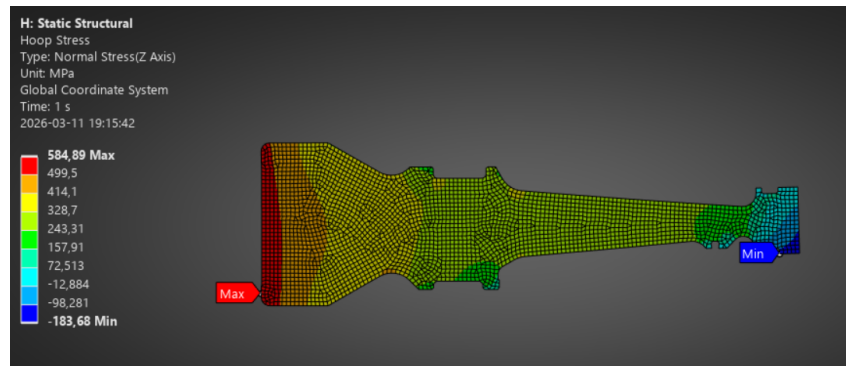


Figure 7: Tangential stress variation due to both the mechanical loads and the temperature variation.

Discussion of Results: The combined tangential stress at the bore is **584.89 MPa**. Comparing this to the purely mechanical stress of the detailed geometry (**477.86 MPa**), we observe that the thermal gradient *increases* the total stress at the inner radius.

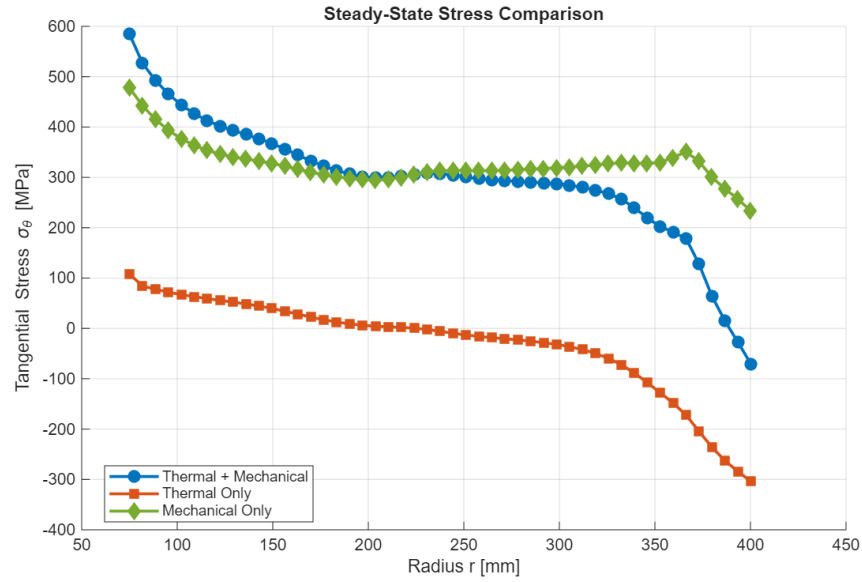


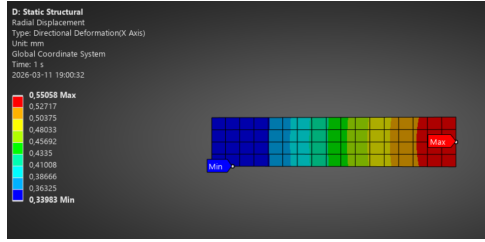
Figure 8: Comparison of tangential stress variations (σ_θ) along the radial coordinate: Thermal only vs. Mechanical only vs. Combined Thermo-Mechanical.

5. Structural Response: Displacements and Strains

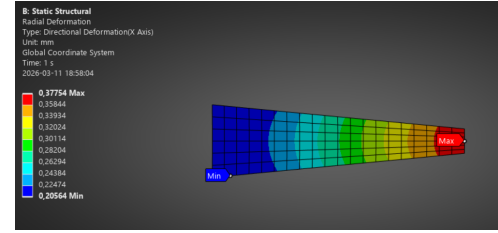
The structural response is governed by the mechanical loads and the internal "clamping" forces caused by non-uniform thermal expansion.

5.1 Radial Displacement Analysis

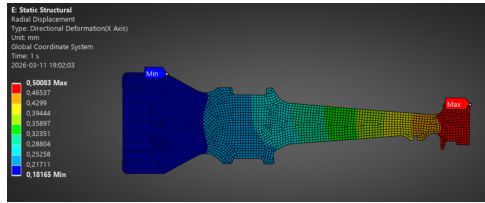
Radial displacement u_r is a result of the centrifugal "stretch" and the thermal expansion. The displacement is highest at the rim due to the cumulative expansion of the inner material.



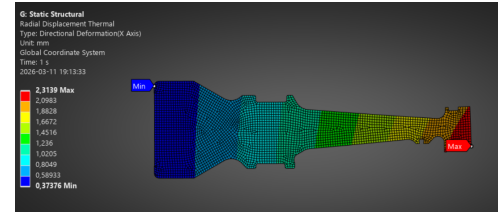
(a) Radial Displacement due Mechanical load in Constant thickness model



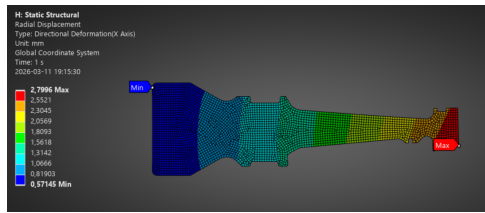
(b) Radial Displacement due Mechanical load in Tapered model



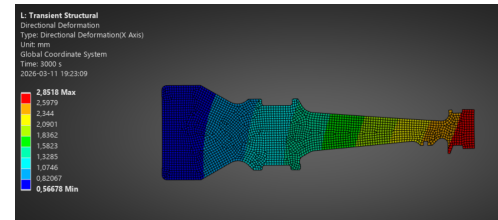
(c) Radial Displacement due Mechanical load in Detailed Model



(d) Radial Displacement due thermal load



(e) Radial Displacement due both thermal and mechanical load

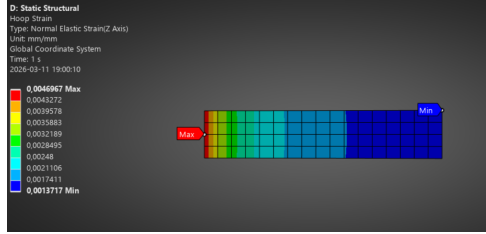


(f) Radial Displacement due both thermal and mechanical load under Transient State

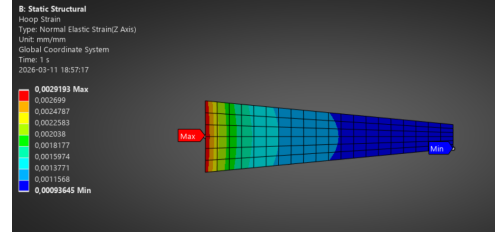
Figure 9: Comparison of Radial Displacement distributions across the three levels of geometric approximation at both steady-state and transient state.

5.2 Strain Components

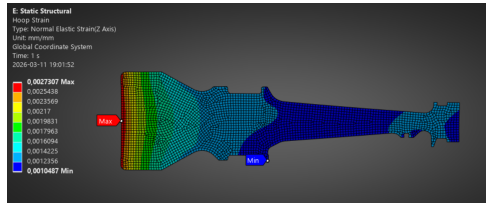
The total strain ε includes the thermal strain ($\alpha\Delta T$) and the mechanical strain. Since the rim is exposed to a higher temperature, it is expected to expand more readily. However, this expansion is restricted by the cooler, stiffer material in the bore. The outer rim, therefore, is in a state of compression, while the inner material is under considerable strain.



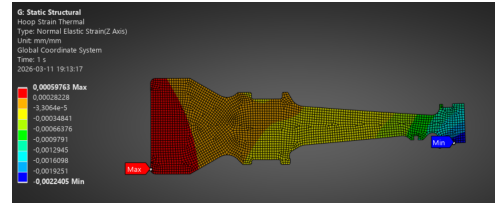
(a) Hoop Strain due Mechanical load in Constant thickness model



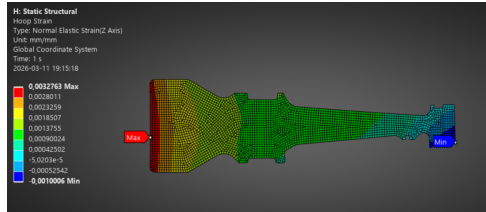
(b) Hoop Strain due Mechanical load in Tapered model



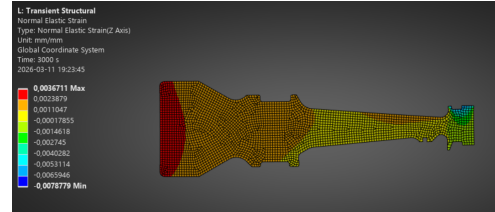
(c) Hoop Strain due Mechanical load in Detailed Model



(d) Hoop Strain due thermal load



(e) Hoop Strain due both thermal and mechanical load

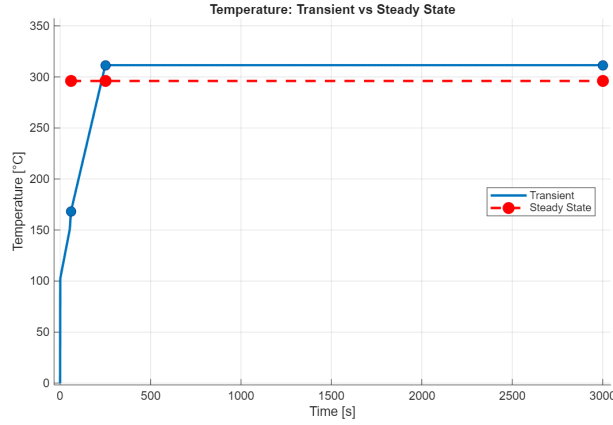


(f) Hoop Strain due both thermal and mechanical load under Transient State

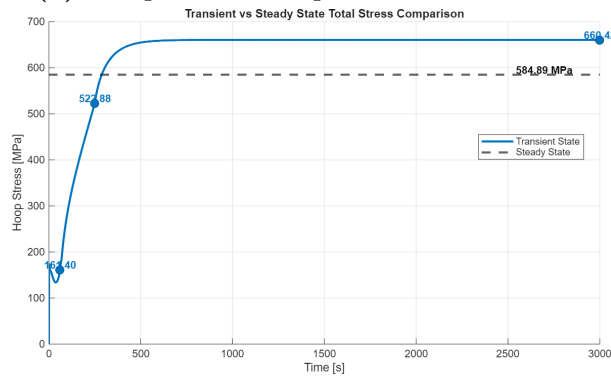
Figure 10: Total tangential strain distribution at $t = 3000$ s.

6. Transient Stress Results and Discussion

The primary conclusion of the study is derived from the tangential stress history at the innermost radius.



(a) Temperature comparison at inner radius

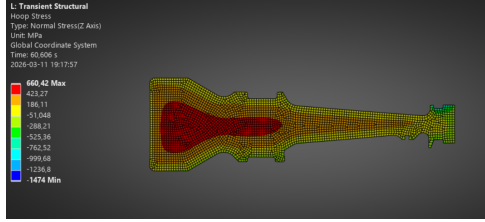


(b) Total Stress Comparison at inner radius

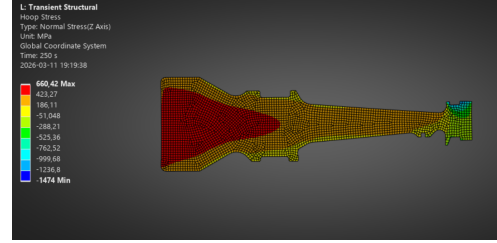
Figure 11: Transient vs Steady State comparison for Temperature and Total Stress at Inner Radius.

6.1 Quasi-Static Superposition

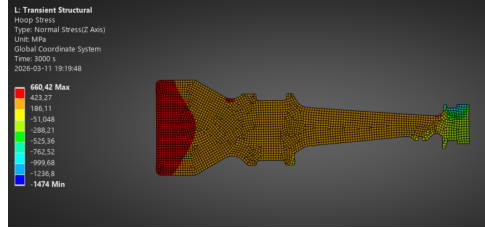
Stresses are calculated at specific time intervals. During each time interval, the centrifugal force, as determined by the ramping rate of the RPM, and the thermal expansion, as determined by the temperature field, are combined. This assumes an effectively instantaneous response to the mechanical loading relative to the thermal response.



(a) Hoop Stress at $t = 60s$



(b) Hoop Stress at $t = 250s$



(c) Hoop Stress at $t = 3000s$

Figure 12: Tangential stress distribution at different timestep $t = 60s$, $t = 250s$ & $t = 3000s$.

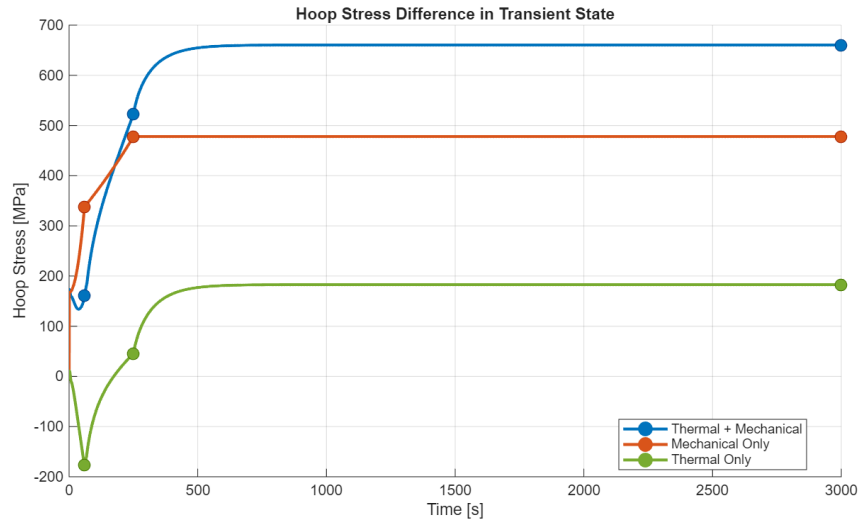


Figure 13: Tangential Stress at the inner radius in transient State.

7. Conclusion

This numerical analysis on the GT35C turbine disc confirms that a steady-state analysis is not sufficient to ensure safety in a component. The transient thermal lag experienced during the start-up phase results in a maximum tangential stress significantly greater than that in the steady-state condition.

In this analysis, where displacements, strains, and temperature are evaluated simultaneously, effects of thermo-mechanical coupling are noted during operation. The outer radius, being in a higher temperature state than the inner radius, will expand more than the inner radius. This will cause a greater stress in the inner radius due to this resistance to expansion. This results in a state of high stress in the inner radius of the disc.

From this analysis on transient heat transfer and quasi-static structural response, it is noted that the structural integrity of this disc is dependent on the temperature. Possible design improvements could involve attempting to reduce this stress by optimizing the start-up phase or by modifying the disc's thermal mass.